

Summer Holiday Worksheet

Science WS-2

1) The laws of refraction (Plane Law & Snell's Law) are:

- * The incident ray, the refracted ray and the normal to the interface at the point of incidence all lie in the same plane.
- * The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media.

2) The image formed by a plane mirror has the following characteristics:

- * Virtual
- * Erect
- * Laterally inverted
- * Same size and distance with respect to the object

3) Focal length $\rightarrow -20$ cm (concave mirror)

Image distance $\rightarrow -30$ cm ; Object height $\rightarrow 4$ cm

By using mirror formula, $\left(\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \right)$

$$\frac{1}{-20} = \frac{1}{-30} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{u} = \frac{1}{-20} - \frac{1}{-30} = -\frac{1}{20} + \frac{1}{30} = \frac{-3+2}{60} = -\frac{1}{60}$$

So, $u = -60 \text{ cm}$

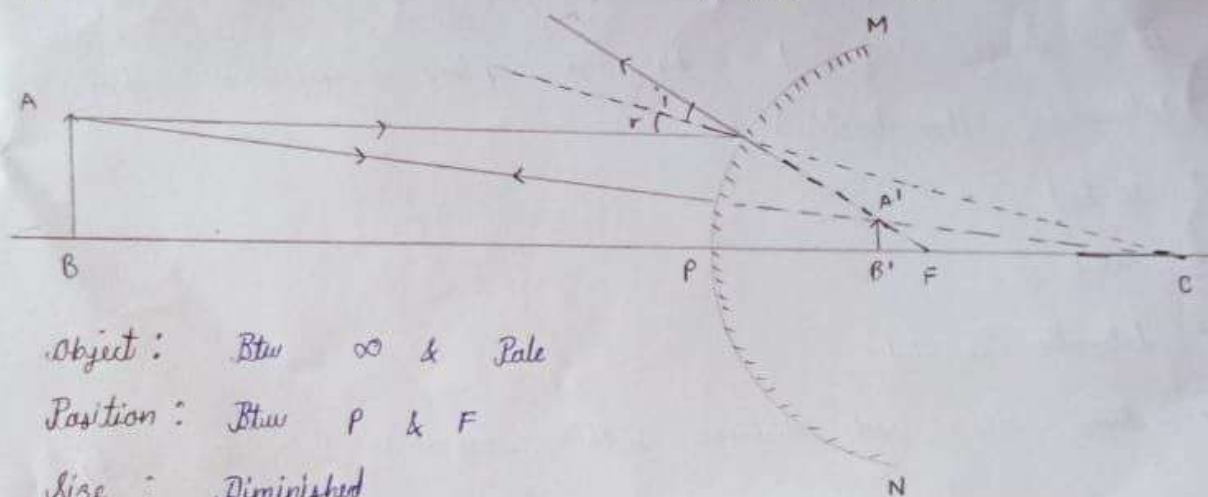
The object should be placed 60 cm in front of the mirror.

Now, $\frac{h'}{h} = \frac{v}{u}$

$\frac{h'}{42} = \frac{-30}{-602} \Rightarrow h' = \frac{11}{2} \text{ cm}$

Image height = 2 cm

4) The type of the mirror as per as the given characteristics is a convex mirror.



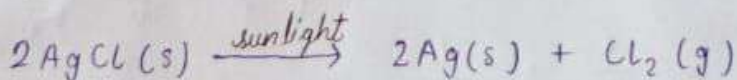
Object: Btw ∞ & Pole

Position: Btw P & F

Size: Diminished

Nature: Virtual and Erect

5) When silver chloride (AgCl) is exposed to sunlight, it decomposes into silver and chlorine.



Type of reaction: Photochemical decomposition reaction

6) a) Phenomenon : Rusting

b) Chemical equation : $4\text{Fe} + 3\text{O}_2 + 6\text{H}_2\text{O} \longrightarrow 4\text{Fe}(\text{OH})_3$

c) Prevention method : Painting, Galvanisation, Alloying, etc.

7) a) The gas evolved when zinc reacts with dilute sulphuric acid is hydrogen gas (H_2).

b) Balanced chemical equation:



c) Hydrogen gas is collected by downward displacement of water because hydrogen is insoluble in water and lighter than air, so it rises and can be collected easily.

8) If platelets were absent, blood would not clot, even from minor injuries. This would lead to excessive bleeding, which could be life threatening.

9) In fishes, the blood circulation is single and simple. Blood passes through the heart only once during one complete cycle of circulation. The heart of the fish has two chambers - one atrium and one ventricle. Blood flows from the heart to the gills for oxygenation and then directly to the rest of the body. In human beings, the blood circulation is double. Blood passes through the heart twice during

one complete cycle. The human heart has four chambers, two atrium and two ventricles. Oxygen poor blood goes from the body to the right side of the heart, then to the lungs for oxygenation, and returns to the left side of the heart before being pumped to the rest of the body.

Thus, circulation in fishes is different than from humans.

- 10) a) The rings of cartilage in the trachea keeps it open and prevents it from collapsing, ensuring that air can pass freely to and from the lungs during breathing.
- b) Herbivores eat plant-based food, which contains cellulose that is harder to digest. A longer small intestine provides more time and surface area for the digestion and absorption of nutrients from this complex food.
- Carnivores on the other hand consume meat which is easier to digest, so they have shorter small intestines.
